VIRTUAL SCHOOL DEVICE MANAGEMENT

New Android Device Setup Guide

Build, install, enroll, verify, and assign a school-owned Android tablet using the current Virtual School pilot environment.

APK	virtual-school-launcher-0.1.0-debug.apk
Package	com.virtualschool.launcher
Android	Minimum 8.0 (API 26) / target API 35
Audience	IT administrators and pilot technicians
Guide date	14 August 2026

Choose the correct path

Use Path A for local evaluation with USB/ADB. Use Path B for school-owned managed enrollment. The portal QR now contains the live Google Android Management API setup-wizard payload.

Pilot build only

This APK is signed with the Android debug certificate. Do not publish it, distribute it broadly, or use it as a production management app.

1. Before you begin

Confirm the device, account, network, and enrollment path before touching a tablet.

Required items

- A school-owned Android 8.0 or newer tablet that can be factory-reset.
- A reliable Wi-Fi network with internet access and no captive portal during enrollment.
- An IT administrator account for the correct school tenant.
- For Path A: a Windows computer with Android platform-tools and a USB data cable.
- For Path B: an active Android Enterprise provisioning QR issued by the portal.

Which path should I use?

Goal	Use	Result
Evaluate the launcher on one tablet	Path A - ADB install	Installable debug launcher; not fully managed
Enroll a school-owned managed tablet	Path B - QR provisioning	Managed device enrollment and policy
Publish to staff or students	Neither yet	Requires a release-signed launcher

Factory reset warning

Managed QR enrollment starts from the Android setup wizard. A factory reset permanently erases apps, accounts, photos, and other local data. Back up approved data and confirm device ownership before resetting.

APK build record

Property	Verified value
File	virtual-school-launcher-0.1.0-debug.apk
Package	com.virtualschool.launcher
Signature	Android Debug certificate; APK Signature Scheme v2
Purpose	Pilot evaluation only

2. Path A - install the pilot APK with ADB

Local evaluation only

ADB installation lets you inspect the launcher UI. It does not enroll the tablet into Android Enterprise, enforce device-owner controls, or apply the server policy automatically.

1

Enable developer access

On the test tablet, open Settings, enable Developer options, then enable USB debugging. Menu names vary by manufacturer.

2

Connect the tablet

Connect with a USB data cable. Accept the RSA debugging prompt only if the computer is trusted. Confirm visibility with `adb devices`.

3

Install or update the launcher

Open PowerShell in the APK folder and run `adb install -r "virtual-school-launcher-0.1.0-debug.apk"`.

4

Open and select the launcher

Launch Virtual School. If Android asks for a Home app, select Virtual School for this evaluation.

5

Confirm basic behavior

Verify the label appears, the launcher opens without a crash, HTTPS shortcuts are allowed, and invalid shortcuts fail closed.

6

Remove the pilot build if required

Run `adb uninstall com.virtualschool.launcher`, or remove the app from Android Settings.

If installation is blocked

Confirm the tablet appears as device, not unauthorized, in adb devices. Reconnect the cable, approve the tablet prompt, and remove any build signed with an incompatible certificate.

3. Path B - create the enrollment batch

1

Sign in as the school's IT administrator

The system derives name, role, and school from the account. Confirm the header shows the intended school.

2

Open Enrollment

Choose Enrollment in the left navigation. Only roles with device enrollment permission can access this screen.

3

Create a controlled batch

Enter a meaningful name, realistic device cap, and expiry. Use a separate batch for each rollout group.

4

Download the QR and provisioning kit

The backend requests a live token from Google. Download the resulting QR image and kit, and store both as sensitive enrollment material.

5

Check status before scanning

Use only an ACTIVE, unexpired batch for the correct school. REVOKED, EXPIRED, or capacity-complete batches must not be reused.

6

Revoke unused access

Select Revoke when a rollout ends early, a QR is exposed, or the batch is no longer required. Revocation also deletes the provider token.

Tenant and token safety

Enrollment lists and actions require device enrollment permission and remain school-scoped. Treat every QR as a credential: do not email it broadly, reuse it across schools, or place it in public documentation.

4. Prepare and provision the tablet

Live Google Android Management provisioning

The portal requests a live enrollment token from the configured Android Management enterprise and encodes Google's exact setup-wizard JSON in the QR. Use only active, unexpired batches created for the correct school tenant.

Provision the school-owned tablet

1. **Confirm the approved model**

   Check the Android device compatibility matrix and verify the tablet supports fully managed, school-owned enrollment.

2. **Factory-reset the device**

   Back up approved data, remove any reset-protection account required by school procedure, and reset to the welcome screen.

3. **Start QR provisioning**

   On supported Android devices, tap the welcome screen six times to open QR setup, then connect to Wi-Fi. Manufacturer behavior can differ.

4. **Scan the portal-issued QR**

   Scan the live QR for the correct school and batch. Never scan a QR received through an untrusted channel.

5. **Allow management setup to finish**

   Keep the device powered and online while the management agent, baseline policy, and approved applications are applied.

Production deployment rule

Deploy a release-signed launcher through managed Google Play or the approved Android Enterprise policy. Do not sideload this debug APK onto production or student devices.

5. Verify, assign, and apply policy

1. **Wait for the first check-in**

   The new tablet should appear in Devices after Android Management and the device integration report its enrollment identity.

2. **Confirm quarantine**

   A newly enrolled, unassigned device should remain in PENDING quarantine. Do not hand it to a student while unassigned.

3. **Match the physical tablet**

   Compare serial number, manufacturer, model, and enrollment record with the label or Settings page on the physical tablet.

4. **Assign the tablet**

   Assign the device to the correct campus and, where applicable, student, class, or device group. Verify the school header before saving.

5. **Apply and verify the baseline policy**

   Confirm the expected launcher, secure browser, approved apps, web controls, and restrictions arrive.

6. **Check compliance and audit history**

   Verify a recent healthy check-in, expected compliance state, and an auditable enrollment and assignment trail.

Stop on a tenant mismatch	If the portal, QR batch, device record, or assignment shows the wrong school, stop immediately. Revoke the batch if necessary and escalate with the device serial number and visible error.

6. Acceptance checklist and troubleshooting

Do not hand over the tablet until every applicable check passes.

Check	Expected result
Identity	Serial/model match the physical school-owned tablet
Tenant	Portal header and inventory record show the correct school
Enrollment	Device appears once; no conflicting enrollment identity
Assignment	Correct campus and student/class/group are recorded
Launcher	Virtual School opens as the intended Home experience
Policy	Approved apps, HTTPS shortcuts, and restrictions are present
Health	Recent check-in, healthy state, and no critical alerts
Security	USB debugging is disabled after local evaluation, if used

Common problems

Symptom	Action
ADB shows unauthorized	Unlock tablet, approve the computer's RSA key, and reconnect USB
APK update fails	Remove an older build signed with a different key, then reinstall
QR is rejected	Confirm the batch is active, unexpired, and for the correct school; create a fresh batch if necessary
Device never appears	Check internet, provider status, check-in integration, and batch validity
Wrong school appears	Sign out immediately; verify staff account and tenant assignment
Device remains pending	Complete assignment and confirm the policy target includes the device

Escalate rather than bypass

Do not disable Android security, reuse another school's QR, expose an enrollment token, or force a device out of quarantine. Record serial number, school, batch, time, and visible error, then escalate.

Project references

- docs/operations/android-enrollment.md - live enrollment lifecycle
- docs/operations/android-launcher.md - launcher behavior and managed configuration
- docs/operations/android-device-compatibility.md - approved-device verification